General Characteristics

201. The atomic radii of the elements are almost same of which series
(a) $Fe - Co - Ni$
(b) $Na - K - Rb$
(c) $F - Cl - Br$
(d) $Li - Be - B$

202. In human body if necessary, the plate, screw or wire used for surgery are made up of
(a) Ni (b) Au
(c) Pt (d) Ta

203. Manganese is related to which block of periodic table
(a) *s*-block (b) *p*-block
(c) *d*-block (d) *f* -block

204. A hard and resistant metal (alloy) generally used in tip of nib of fountain pen is
(a) $Os. Ir$ (b) $Pt. Cr$
(c) $V. Fe$ (d) $Fe. Cr$

205. Chloride of which of the following elements will be coloured
(a) Silver (b) Mercury
(c) Zinc (d) Cobalt

206. Which of the following ions has the highest magnetic moment
(a) Ti^{3+} (b) Sc^{3+}
(c) Mn^{2+} (d) Zn^{2+}

207. Cerium $(Z = 58)$ is an important member of the lanthanoids. Which of the following statements about cerium is
(a) The +4 oxidation state of cerium is not known in solutions
(b) The +3 oxidation state of cerium is more stable than the +4 oxidation state
(c) The common oxidation states of cerium are +3 and +4
(d) Cerium (IV) acts as an oxidizing agent

208. Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them
(a) $(n-1)d^3ns^2$
(b) $(n-1)d^5ns^1$
(c) $(n-1)d^8ns^2$
(d) $(n-1)d^5ns^2$

209. Among the following series of transition metal ions, the one where all metals ions have $3d^2$ electronic configuration is
(a) $Ti^{4+},\ V^{3+},\ Cr^{2+},\ Mn^{3+}$
(b) $Ti^{2+},\ V^{3+},\ Cr^{4+},\ Mn^{5+}$
(c) $Ti^{3+},\ V^{2+},\ Cr^{3+},\ Mn^{4+}$
(d) $Ti^{+},\ V^{4+},\ Cr^{6+},\ Mn^{7+}$

210. Lanthanoids are
(a) 14 elements in the sixth period (atomic no. = 58 to 71) that are filling 4*f* sublevel
(b) 14 elements in the seventh period (atomic no. = 58 to 71) that are filling 4*f* sublevel

(c) 14 elements in the sixth period (atomic no. = 90 to 103) that are filling 4*f* sublevel

(d) 14 elements in the seventh period (atomic no. = 90 to 103) that are filling 4*f* sublevel

211. Which of the following metals make the most efficient catalyst

(a) Transition

(b) Alkali

(c) Alkaline earth

(d) Coloured metals

212. Lanthanides and actinides resemble

(a) Electronic configuration

(b) Oxidation state

(c) Ionization energy

(d) Formation of complexes

213. The lanthanide contraction relates to

(a) Atomic radii

(b) Atomic as well as M^{3+} radii

(c) Valence electrons

(d) Oxidation states

(e) Ionisation energy

214. Which of the following species is expected to show the highest magnetic moment? (At. Nos.: *Cr*=24, *Mn* = 25, *Co* = 27, *Ni* = 28, *Cu* = 29)

(a) Cr^{2+} (b) Mn^{2+}

(c) Cu^{2+} (d) Co^{2+}

(e) Ni^{2+}

215. Which one belongs to 3*d*-transition

(a) Copper (b) Gold

(c) Cobalt (d) Silver

216. Which one of the following organisation's iron and steel plant was built to use charcoal as a source of power, to start with, but later switched over to hydroelectricity

(a) The Tata Iron and Steel Company

(b) The Indian Iron and Steel Company

(c) Mysore Iron and Steel Limited

(d) Hindustan Steel Limited

217. Which of the following is the correct sequence of atomic weights of given elements

(a) $Fe > Co > Ni$

(b) $Ni > Co > Fe$

(c) $Co > Ni > Fe$

(d) $Fe > Ni > Co$

218. Which of the following element has maximum first ionisation potential

(a) *V* (b) *Ti*

(c) *Cr* (d) *Mn*

219. A metal *M* having electronic configuration

M -$1s^2 2s^2 2p^6 3s^2 3p^6 3p^6 3d^{10} 4s^1$

(a) *s*-block element

(b) *d*-block element

(c) *p*-block element

(d) None of these

220. Identify the transition element

(a) $1s^2,\ 2s^2 2p^6,\ 3s^2,\ 3p^6,\ 4s^2$

(b) $1s^2,\ 2s^2 2p^6,\ 3s^2,\ 3p^6 3d^2,\ 4s^2$

(c) $1s^2,\ 2s^2 2p^6,\ 3s^2,\ 3p^6 3d^{10},\ 4s^2 4p^2$

(d) $1s^2,\ 2s^2 2p^6,\ 3s^2,\ 3p^6 3d^{10},\ 4s^2 4p^1$